CompTIA Network+ (N10-009)

Chapter 1: The OSI Model Explained

A deep dive into the 7-layer framework every network engineer speaks fluently — and why you need to too.

A Study Guide for Aspiring Network Professionals

By Hassen Hannachi | N10-009 Study Series | March 2026

> *"Before you can troubleshoot a network, you need a shared language. The OSI model is that language — a seven-layer map that lets engineers pinpoint exactly where something breaks."*

As I work through my Network+ N10-009 exam prep, Chapter 1 sets the stage for everything that follows. Without understanding the OSI model, conversations about routing, switching, firewalls, and protocols become noise. With it, they become precise and actionable.

This post is my study breakdown — written to solidify my own understanding and hopefully help others on the same path.

What Is the OSI Model?

The **Open System Interconnection (OSI) Reference Model** was developed by the **International Organization for Standardization (ISO)** in 1984. It's a conceptual framework that standardizes how different network devices and protocols communicate with each other — regardless of the vendor or underlying technology.

Think of it as a universal blueprint. When two devices from completely different manufacturers need to talk, the OSI model provides the rulebook that makes that possible.

> **Key Point:** The OSI model is conceptual, not a literal implementation. It describes what should happen at each stage of network communication — not necessarily how a specific protocol does it.

Understanding these layers helps with:

- Troubleshooting network issues
- Identifying device functions
- Understanding security boundaries
-

The model divides network communication into **7 distinct layers**, each with a specific role. Data travels down these layers on the sending side and back up on the receiving side — a process called **encapsulation** and **decapsulation**.

OSI Model — All 7 Layers at a Glance

#	Layer Name	PDU	Key Protocols / Devices
7	Application	Data	HTTP, HTTPS, FTP, SMTP, DNS, DHCP, SSH
6	Presentation	Data	SSL/TLS, JPEG, ASCII, Encryption
5	Session	Data	NetBIOS, PPTP, RPC, SIP
4	Transport	Segment / Datagram	TCP, UDP — Ports & reliability
3	Network	Packet	IP, ICMP, Routers, IPv4/IPv6
2	Data Link	Frame	Ethernet, MAC, Switches, ARP
1	Physical	Bits	Cables, Hubs, NIC, Fiber, Wi-Fi

🧠 Memory Aid — Bottom to Top (Layer 1 → 7)

```
P  →  Please     (Physical)
D  →  Do         (Data Link)
N  →  Not        (Network)
T  →  Throw      (Transport)
S  →  Sausage    (Session)
P  →  Pizza      (Presentation)
A  →  Away       (Application)
```

The 7 Layers — Deep Dive

Layer 1 — Physical

The Physical layer is the foundation. It deals with the raw transmission of bits — electrical signals, light pulses, or radio waves — over a physical medium. This layer doesn't care about the meaning of the data, only how to send 0s and 1s from one point to another.

Devices: Cables (copper, fiber), Hubs, Repeaters, Network Interface Cards (NICs), wireless access points at the signal level.

Real-world example: When your Ethernet cable carries voltage changes representing bits — that's Layer 1 in action. If this layer fails, nothing else works. Always check your cables first.

Layer 2 — Data Link

The Data Link layer takes raw bits and organizes them into frames. It handles node-to-node communication on the same network segment, using MAC (Media Access Control) addresses to identify devices.

It has two sub-layers: LLC (Logical Link Control) — handles flow control and error checking — and MAC — handles physical addressing and media access.

Devices: Switches, Bridges, NICs (MAC address level).

Protocols: Ethernet, Wi-Fi (802.11), ARP (Address Resolution Protocol).

Key concept: Switches operate at Layer 2. They learn which MAC addresses are on which ports and forward frames intelligently — unlike Layer 1 hubs that just broadcast everything.

Layer 3 — Network

The Network layer is responsible for logical addressing and routing — determining the best path to get a packet from source to destination across multiple networks. This is where IP addresses live.

Devices: Routers, Layer 3 Switches.

Protocols: IPv4, IPv6, ICMP, OSPF, BGP, RIP.

Key concept: Routers operate at Layer 3. When your data needs to cross the internet, routers read IP addresses and make forwarding decisions at every hop.

Layer 4 — Transport

The Transport layer is the workhorse of reliable communication. It breaks data into segments (TCP) or datagrams (UDP), handles error recovery, flow control, and reassembly.

The two main protocols here are:

- TCP (Transmission Control Protocol) — Connection-oriented, reliable, ordered delivery. Used for HTTPS, email, file transfers.
- UDP (User Datagram Protocol) — Connectionless, fast, no guarantee. Used for streaming, gaming, DNS, VoIP.

> **Exam Tip:** Know your port numbers! TCP/UDP, port numbers, and the difference between connection-oriented vs. connectionless communication are heavily tested on the Network+ exam.

Layer 5 — Session

The Session layer manages sessions — the logical connections between applications. It establishes, maintains, and terminates communication sessions between two devices.

Protocols: NetBIOS, PPTP, RPC, SIP, NFS.

Real-world example: When you log in to a web application, the session layer tracks that conversation so you don’t have to re-authenticate on every click.

Layer 6 — Presentation

The Presentation layer is the translator. It ensures data from the application layer is in a format the receiving system can understand. It handles data encoding, compression, and encryption/decryption.

Functions: Character encoding (ASCII, Unicode), data compression (JPEG, MP3, ZIP), encryption/decryption (SSL/TLS).

Key concept: When your browser uses HTTPS, the encryption/decryption process lives conceptually at Layer 6. The padlock icon? That’s Presentation layer work.

Layer 7 — Application

The Application layer is the closest to the end user. It provides network services directly to user applications. This is NOT the application itself (like Chrome or Outlook) — it's the interface that allows applications to use network services.

Protocols: HTTP/HTTPS, FTP, SMTP, POP3, IMAP, DNS, DHCP, SNMP, Telnet, SSH.

Key concept: When you type a URL in your browser, DNS resolution, HTTP requests — all of that is initiated at Layer 7.

Why Do We Use the OSI Model?

If the OSI model is conceptual and not directly implemented, why do we bother? Three powerful reasons:

1. Troubleshooting

"Is this a Layer 1 (physical) problem or a Layer 3 (routing) problem?" The model gives you a structured, systematic approach to diagnose issues from the ground up. Start at Layer 1 and work your way up.

2. Vendor Interoperability

Different vendors' devices can communicate because they all adhere to the same layer standards. A Cisco switch can talk to a Juniper router because they both speak the same layered language.

3. Standardized Development

Protocol designers and software developers use the OSI model as a reference, so new technologies integrate seamlessly into existing networks.

The TCP/IP Model

While the OSI model is the reference framework, the **TCP/IP model** (also called the DoD model) is what the internet actually runs on. It collapses OSI's 7 layers into **4 layers**.

The TCP/IP model is more commonly used in real-world networking, but the OSI model is more detailed and better for learning and troubleshooting.

TCP/IP Layers → OSI Mapping

#	TCP/IP Layer	Corresponding OSI Layers	Key Protocols
4	Application	OSI 5 (Session) + 6 (Presentation) + 7 (Application)	HTTP, FTP, DNS, SMTP, SSH, DHCP
3	Transport	OSI 4 (Transport)	TCP, UDP
2	Internet	OSI 3 (Network)	IP, ICMP, ARP, OSPF, BGP
1	Network Access	OSI 1 (Physical) + 2 (Data Link)	Ethernet, Wi-Fi, MAC, NIC drivers

TCP/IP vs. OSI — Key Differences

Aspect	OSI Model	TCP/IP Model
Layers	7 layers	4 layers
Origin	ISO (1984) — theoretical	DARPA (1970s) — practical
Purpose	Reference / teaching model	Actual internet protocol suite
Session & Presentation	Separate dedicated layers	Merged into Application layer
Physical & Data Link	Two separate layers	Combined into Network Access
Real-world use	Troubleshooting, education	Actual internet communication
Protocol-specific?	Protocol-independent	Built around TCP and IP

> **Exam Perspective:** The Network+ exam uses both models. You'll need to know which OSI layers map to which TCP/IP layers, which devices operate at which layer, and which protocols belong where.

Chapter 1 — Summary Takeaways

- The OSI model has 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
- Each layer has a specific PDU (Protocol Data Unit): Bits → Frames → Packets → Segments → Data.
- Mnemonic: "Please Do Not Throw Sausage Pizza Away" (Layer 1 → 7).
- The OSI model is used for troubleshooting, interoperability, and standardization.
- The TCP/IP model is the real-world implementation — 4 layers that map to OSI's 7.
- Key devices: Hubs (L1), Switches (L2), Routers (L3), Firewalls (L3/L4).
- Know your protocols per layer — HTTP, TCP, IP, Ethernet, etc.

That's Chapter 1 wrapped up. The OSI model feels abstract at first, but once you start layering concepts on top of it (routing, switching, firewalls, VPNs), you'll appreciate having this mental model. It's the foundation everything else is built on.

Next up: **Chapter 2 — Networking Appliances, Applications, and Functions**. Stay tuned! 🚀